

Understanding Fractions

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A fraction names part of a whole. The **numerator** (top number) tells how many **parts you have**. The **denominator** (bottom number) tells how many **equal parts** are in the whole. The parts must be equal. A **bigger denominator** means the whole is cut into more pieces, so each piece is **smaller**.
Example: in $\frac{3}{4}$, you have 3 of the 4 equal parts.

Round 1 — Retrieve It

1. What does the numerator (top number) tell you?

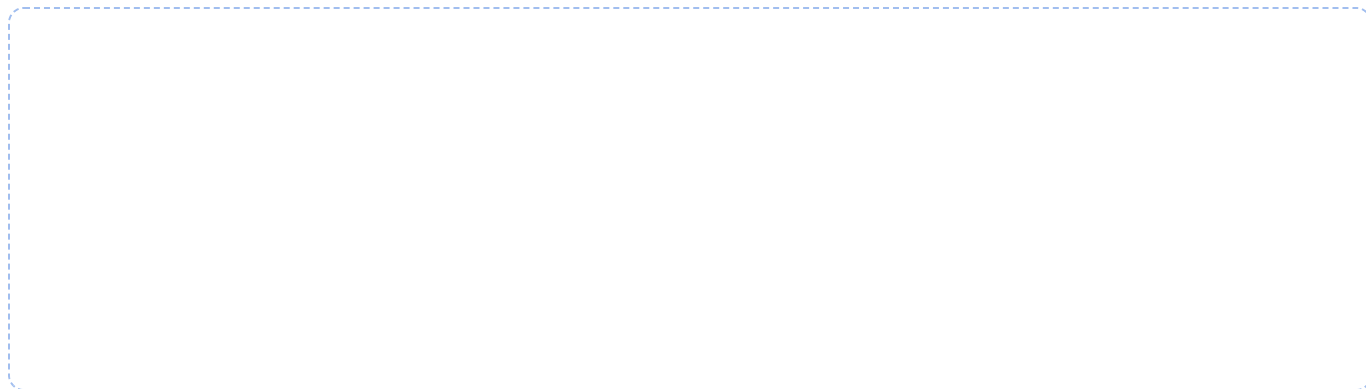
2. What does the denominator (bottom number) tell you?

3. Do the parts of a whole have to be equal?

4. A bigger denominator makes each piece bigger or smaller?

Round 2 — Sketch It

Draw two rectangles the same size. Split one into fourths and shade $\frac{3}{4}$. Split the other into eighths and shade $\frac{3}{8}$. Label each fraction.



Understanding Fractions — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A pizza is cut into 6 equal slices and you eat 2. Write the fraction you ate and name the numerator and denominator.

2. Which is a bigger piece, $\frac{1}{3}$ of a bar or $\frac{1}{6}$ of the same bar? Explain.

3. A shape is split into 4 unequal parts. Can you call each part $\frac{1}{4}$? Why or why not?

Round 4 — Reason from Evidence

Kim says $\frac{1}{8}$ is bigger than $\frac{1}{4}$ because 8 is bigger than 4. Is she correct? Use what you know about denominators and piece size to explain.

Understanding Fractions — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain what the numerator and denominator each tell you. Use the words parts, equal, and whole.

C**Connect**

Connect fractions to sharing food. Tell how you would split one sandwich equally among 4 people and name each share.

E**Extend**

Draw or describe a whole cut into eighths. Explain why each eighth is smaller than each fourth of the same whole.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Understanding Fractions

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. How many parts you have.
2. How many equal parts are in the whole.
3. Yes, the parts must be equal.
4. Smaller (more pieces means each piece is smaller).

Sketch It — look for

Two same-size rectangles. One in fourths with 3 parts shaded ($\frac{3}{4}$); one in eighths with 3 parts shaded ($\frac{3}{8}$). $\frac{3}{4}$ covers more area than $\frac{3}{8}$.

Apply It

1. $\frac{2}{6}$. Numerator 2 (parts eaten), denominator 6 (equal slices in the whole).
2. $\frac{1}{3}$ is bigger. The bar is cut into fewer, larger pieces, so each third is bigger than each sixth.
3. No. The parts must be equal to be called $\frac{1}{4}$. Unequal parts are not fourths.

Reason from Evidence — look for

No. A bigger denominator means more, smaller pieces. $\frac{1}{8}$ splits the whole into 8 pieces, so each is smaller than $\frac{1}{4}$. So $\frac{1}{4} > \frac{1}{8}$.

ACE — Extend look-for

A whole shown in 8 equal parts. Student explains that cutting into more equal parts makes each part smaller, so $\frac{1}{8} < \frac{1}{4}$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.